# Do Bulls and Bears Move Across Borders? The t-0.5 Framework and International Information Transmission to Asia-Pacific Markets

Ryo Kinoshita

*Graduate School of Economics, Kobe University*

## Abstract

This study measures the extent to which information available at the close of the New York market (06:00 Japan Standard Time, JST) predicts the direction of Asian markets.

To this end, the study introduces the "t-0.5 framework." Defining the New York close as the origin of information generation (t=0) and the close of the same market on the following trading day as the end of a full cycle (t=1), the Tokyo open occupies an intermediate point in this cycle (t≈0.5). Under full information efficiency, the directional accuracy (DA) at t=0.5 should converge to 50%, and this study evaluates the deviation from this level as a measure of the residual degree of information transmission. We interpret the excess over 50% as the magnitude of the predictability that information at the New York close holds for the next-day rise or fall of Asian equity indices.

Using a walk-forward logistic regression with a 2,800-day training window and 522 out-of-sample (OOS) windows over 2021–2023, the Nikkei 225 attains DA = 67.82% ($Z = 8.141$, $p < 0.001$), exceeding the random-walk benchmark by 17.82 percentage points in directional predictability. A zero-parameter VIX-sign rule (predicting an Asian-market rise when VIX falls) also achieves DA = 69.35%, nearly matching the logistic regression.

A carrier screening using a SHAP analysis expanded to ten variables identifies the S&P 500 return (SPX, 0.431) as the largest carrier, followed by VIX (0.256) and TLT, the long-term U.S. Treasury ETF (0.120). VIX appeared dominant in the base four-variable model because its strong negative correlation with SPX made it a proxy for the direction of SPX. SPX, VIX, and TLT—reflecting market direction, investor sentiment, and safe-haven demand, respectively—are referred to here, for convenience, as the "NY equity-risk complex." A prediction-horizon analysis shows that the signal collapses to DA = 49.90% one trading day later (h=2), statistically indistinguishable from a random walk. KOSPI (60.94%***), the Hang Seng (58.39%***), and ASX 200 (67.28%***) all achieve significant DA, and significance is maintained across all sub-periods (2021–2023). Predictability decays rapidly within one business day, with consistent results across the four markets and across different market environments.

This study makes five contributions: (1) it proposes directional accuracy at t=0.5 as an empirical indicator of information transmission; (2) it confirms out-of-sample performance through walk-forward validation; (3) it demonstrates the effectiveness of a simple baseline, the VIX-sign rule; (4) it identifies SPX, VIX, and TLT as the main carriers via SHAP; and (5) it describes the

heterogeneity of market-specific interest-rate and currency channels across four Asia-Pacific markets.

*Keywords: international information transmission, price discovery, directional accuracy, t-0.5 framework, VIX, walk-forward validation, SHAP, Asia-Pacific markets, market efficiency*

## 1. Introduction

When the New York Stock Exchange closes, its final prices embody the collective assessment of millions of market participants regarding the state of the global economy. Three hours later, the Tokyo Stock Exchange opens. How much of that assessment is already reflected in Tokyo prices, and how much remains unincorporated? This is the central question of this study.

This study operationalizes the question through the t-0.5 framework. Defining t=0 as the New York close and t=1 as the close of the same market on the next trading day (one information cycle), t≈0.5 corresponds to the Tokyo open (09:00 JST), an intermediate point in the global 24-hour trading relay. If markets were fully efficient, information at t=0 would be fully priced worldwide before the next market opens, and directional accuracy (DA) at t=0.5 would converge to 50%. A systematic excess over this 50% benchmark represents the magnitude of the predictability that information at the New York close holds for the direction of next-day Asian index returns. Note that the Nikkei 225 close-to-close return used here contains both the opening gap and the intraday return, so it does not directly measure unpriced information at the open.

Lin, Engle, and Ito (1994) first demonstrated, through intraday GARCH-based signal extraction, that New York close information reaches the Tokyo open, establishing the existence of this transmission channel. However, their framework targeted volatility correlations and could not directly measure how much of the signal remains actionable in real time. Eun and Shim (1989) and Hamao, Masulis, and Ng (1990) reached similar conclusions within VAR and GARCH-M frameworks; this study reframes the question as one of directional accuracy. Out-of-sample evaluation of overnight directional predictability has remained limited.

Using a walk-forward logistic regression with a 2,800-day training window, we obtain DA = 67.82% ($Z = 8.141$, $p < 0.001$) for the Nikkei 225 over 522 OOS windows (2021–2023). Having confirmed that a residual exists, we then identify its carrier. A SHAP decomposition makes the CBOE VIX the dominant predictor (mean|SHAP| = 0.484). The most direct confirmation is that a zero-parameter rule—if VIX falls at the New York close, predict a rise in Tokyo—achieves DA = 69.35%. This is statistically indistinguishable from the full logistic regression, showing that the single binary variable of the sign of VIX captures the essence of the overnight signal.

When a ten-variable carrier screening is conducted, the S&P 500 return (SPX) emerges as the leading carrier, surpassing VIX; VIX is identified as an independent residual-fear channel, and the long-term U.S. Treasury ETF (TLT) as a safe-haven channel. VIX appeared dominant in the base four-variable model because, through its strong negative correlation with SPX, it functioned as a proxy for the direction of SPX.

The second main finding concerns the temporal boundary of transmission. The prediction-horizon analysis shows that the signal collapses to DA = 49.90% at h=2 (the close of the next Asia-Pacific trading day), statistically indistinguishable from a random walk. The information lifespan is exactly one trading day, and this cliff-like decay forms a coherent pattern of statistical significance at t=0.5 and efficiency on the next trading day (h=2).

This study makes five contributions.

First, it proposes directional accuracy at t=0.5 as an operational indicator of the residual degree of international information transmission.

Second, through walk-forward validation with a 2,800-day training window, it confirms statistically significant out-of-sample predictability. Through SHAP analysis, it shows that SPX, VIX, and TLT are especially important features in the predictive model.

Third, it shows that the zero-parameter VIX-sign rule achieves accuracy comparable to the logistic regression, clarifying the economic interpretation of the signal, and that predictability decays rapidly within one business day, clarifying its temporal characteristics.

Fourth, through sub-period analysis over 2021–2023, it shows that the results are robust across the post-COVID recovery, the rapid rate-hike phase, and the stabilization phase, confirming that predictability is maintained under different market environments.

Fifth, comparing four Asia-Pacific markets (Japan, Korea, Hong Kong, and Australia), it describes the heterogeneity of interest-rate and currency channels across markets.

The remainder of the paper is organized as follows. Section 2 reviews prior research, and Section 3 describes the data and methods. Section 4 presents the results, and Section 5 provides discussion. Section 6 concludes, and Section 7 outlines directions for future research.

## 2. Literature Review

### 2.1 International Information Transmission

Research on the co-movement of international equity markets dates back to Grubel (1968), which demonstrated the diversification benefits arising from low cross-country correlations. Eun and Shim (1989) applied VAR analysis to nine major markets (1980–1985), showing that U.S. innovations explain 11–42% of the forecast-error variance of foreign markets within 20 days and that the Japanese market is identified as a pure information receiver. Our results reinforce this characterization.

King and Wadhwani (1990) introduced the concept of contagion, in which mispricing propagates across markets through the inference of rational agents who cannot distinguish common shocks from idiosyncratic ones. Their model explains the sharp rise in cross-market correlations after the 1987 crash and shows theoretically how the imperfection of information strengthens the transmission mechanism. Our t-0.5 framework operationalizes this view: predictability at t=0.5 is precisely a measure of this imperfect incorporation.

Hamao, Masulis, and Ng (1990) applied GARCH-M to daily returns for Tokyo, London, and New York and demonstrated significant volatility spillovers from New York to Tokyo. Lin, Engle, and Ito (1994) refined this analysis using intraday data, showing that the information content of U.S. intraday returns predicts Japan's overnight returns. Their framework established the time-zone asymmetry that motivates the present study. This study extends that work from the detection of volatility spillovers to the real-time measurement of directional accuracy.

The literature since the 1990s has provided a more refined understanding of the speed and channels of information transmission. Karolyi (1995) applied a GARCH-M model to the U.S.–Canada markets and showed that volatility spillovers are not symmetric. Richards (1996) analyzed the co-movement of international equity returns across G7 markets from a long-horizon perspective and presented evidence of cointegrating relationships. All of these studies rely on linear, parametric methods such as VAR and GARCH, which have the potential limitation of being unable to capture nonlinear information-transmission dynamics.

This study differs from this prior research in three respects. First, it targets directional accuracy (DA) rather than volatility correlation. Second, it adopts a strict walk-forward, out-of-sample design, evaluating real-time predictability rather than in-sample fit. Third, through SHAP analysis, it identifies the market channels that carry the signal in an interpretable manner.

## 2.2 Machine Learning and Interpretability in Financial Forecasting

While the application of neural networks to financial forecasting has a long history, rigorous out-of-sample evaluation using walk-forward validation has only recently become standard. Fischer and Krauss (2018) applied an LSTM (long short-term memory; Hochreiter & Schmidhuber, 1997) to S&P 500 constituent returns and demonstrated strong performance in cross-sectional stock selection. Gu, Kelly, and Xiu (2020) provided the most comprehensive comparison of machine-learning methods for stock-return prediction, showing that neural networks consistently outperform linear models.

This study differs from both in its focus on cross-border directional prediction that exploits time zones. Moreover, the finding that a walk-forward LSTM using the same feature set does not outperform logistic regression (DA = 64.51% vs. 67.82%) suggests that, at least under the feature set used here, a relatively simple linear model can achieve sufficiently high predictive performance. No clear improvement from the LSTM's additional nonlinear capacity or sequential memory was observed.

SHAP (SHapley Additive exPlanations; Lundberg & Lee, 2017) is a method for attributing individual predictions to input features, based on the Shapley value from cooperative game theory. For linear models, LinearExplainer computes SHAP values efficiently, consistent with this study's logistic-regression setting. Using SHAP in financial forecasting allows examination not only of predictive accuracy but also of the carriers of the signal; it is widely used to visualize feature contributions to model predictions and to bridge predictive performance and economic interpretation. However, SHAP is a method for explaining feature contributions and does not identify causal relationships among features.

# 3. Data and Methods

## 3.1 Data

Daily closing prices for four variables were obtained from Yahoo Finance from December 2009 to January 2024: the Nikkei 225 (^N225), the CBOE Volatility Index (^VIX), the 10-year U.S. Treasury yield (^TNX), and the USD/JPY exchange rate (USDJPY=X). All variables are converted to log returns. After removing missing values due to holiday mismatches, the final sample for the Nikkei 225 is N = 3,323 trading days. For the cross-market analysis, samples are aligned using only business days on which the common predictors (VIX, TNX, USDJPY) are available (KOSPI: N = 3,354; Hang Seng: N = 3,373; ASX 200: N = 3,455). The six additional variables used in the carrier screening of Section 4.6 (SPX, TLT, GLD, DXY, OIL, and HYG) are obtained from the same source (Yahoo Finance) over the same sample period and are likewise available by the New York close.

The choice of variables reflects the t-0.5 framework. Each variable is available as a predictor at the New York close (06:00 JST), before the Tokyo open (09:00 JST). VIX proxies global risk appetite, TNX the U.S. interest-rate environment, and USDJPY currency-side risk-on/risk-off conditions. The one-day lag of the Nikkei 225 return captures own-market momentum. All explanatory variables are constructed solely from information available at the prediction time, by design avoiding look-ahead bias. Times are stated in U.S. Standard Time; during U.S. daylight saving time the New York close corresponds to 05:00 JST (four hours before the Tokyo open). In either case, all predictors are confirmed before the Tokyo open, preserving the information ordering.

## 3.2 Descriptive Statistics

Table 1 reports descriptive statistics for the daily log returns. The sample period spans about 14 years, from December 1, 2009 to January 1, 2024, and includes the post-global-financial-crisis recovery, the European debt crisis (2011–2012), Brexit (2016), the COVID-19 shock (2020), and the Fed's rapid rate-hike phase (2022), among other phases of substantial change in the market environment. This period encompasses both calm and crisis conditions and is well suited for assessing the robustness of a predictive model.

**Table 1: Descriptive statistics of daily log returns (December 1, 2009 – January 1, 2024)**

| Variable | N | Mean (%) | SD (%) | Min (%) | Max (%) | Skewness | Excess kurtosis |
|---|---|---|---|---|---|---|---|
| Nikkei 225 | 3323 | 0.038 | 1.330 | -11.154 | 7.731 | -0.357 | 4.604 |
| VIX | 3323 | -0.017 | 7.937 | -49.641 | 76.825 | 1.099 | 6.887 |
| 10-year U.S. Treasury yield | 3323 | 0.005 | 3.033 | -38.162 | 40.480 | -0.136 | 31.621 |
| USD/JPY | 3323 | 0.015 | 0.607 | -4.650 | 3.981 | -0.146 | 4.634 |

VIX shows the highest volatility (standard deviation = 7.94%) and positive skewness (1.10), reflecting the asymmetric nature of fear. Although the unconditional correlation between VIX and the Nikkei 225 is small ($r = -0.147$), VIX may contain important information for directional

prediction. The asymmetry whereby sharp rises in VIX cluster in down markets is discussed below. In the predictive analysis, VIX shows a high predictive contribution as a leading explanatory variable, suggesting that it may contain information useful for prediction that simple linear correlation cannot capture.

### 3.3 The t-0.5 Framework

The prediction target is the sign (up/down) of the day-t Nikkei 225 log return, using only information available at the New York close (06:00 JST) on the previous trading day t−1. This design avoids look-ahead bias and ensures temporal consistency between the explanatory and target variables.

The framework is named as follows. Defining t=0 as the New York close (three hours before the Tokyo open) and t=1 as the close of the next trading day (the completion of one information cycle), the Tokyo open corresponds to t≈0.5. Directional accuracy (DA) at this t=0.5 is the fundamental quantity measured.

Note that t in this framework is not a ratio of elapsed physical time but an index representing a conceptual position within the information cycle. Although the Tokyo open is three hours after the New York close, daily data cannot directly observe continuous information absorption within the cycle, so the daily directional accuracy is used as its proxy. We use t=0.5 as a convenient notation for the intermediate stage of the information-transmission process.

### 3.4 Main Model: Logistic Regression

The main model is logistic regression (sklearn.linear_model.LogisticRegression, solver=lbfgs, max_iter=200), with the four variables standardized by a StandardScaler fit on the training window only. There are three reasons for choosing logistic regression rather than an LSTM.

First, a walk-forward LSTM using the same feature set attains only DA = 64.51%, below the logistic regression's 67.82%. Second, logistic regression is deterministic and requires no seed protocol, early stopping, or architecture selection, making it more reproducible than a neural network. Third, for linear models, shap.LinearExplainer enables stable interpretation of feature contributions.

The LSTM used as a comparison benchmark has the following specification: a single LSTM layer with 50 units, Dropout 0.1, a Dense output, the Adam optimizer (learning rate 0.001), a 20-day lookback, and six seeds {0,1,2,3,7,42}. The seed-averaged DA was 64.51% (standard deviation = 0.82%). This indicates that, at least under the feature set used here, no additional predictive improvement from a complex nonlinear model was observed.

### 3.5 Walk-Forward Validation

We adopt a fixed-length (2,800-day) rolling-window walk-forward procedure. At each point, the model is estimated on the prior 2,800 business days of data, and a single out-of-sample prediction is generated for the immediately following business day (STEP=1). This procedure yields 522 independent OOS predictions for the Nikkei 225. To prevent information leakage, the StandardScaler is re-estimated for each training window, and its parameters are used to transform

the corresponding test data. The OOS period is 2021–2023, including the post-COVID recovery, the high-inflation phase, and the monetary-tightening phase in major economies.

### 3.6 Prediction-Horizon Analysis

To examine the lifespan of the t-0.5 signal, we estimate walk-forward logistic regressions for prediction horizons h ∈ {1,2,3,5,10} days ahead. Taking h=1 as the base case, we test whether the information advantage disappears as h increases. The same procedure is applied to each horizon to evaluate the temporal decay of predictability.

Because daily closing data provide only one observation point per information cycle, the within-cycle positions t=0.5 (Tokyo open) and t=1 (next New York close) are collapsed into the same daily observation and cannot be separately estimated. The horizon axis (h, in business days) is distinct from the continuous within-cycle position; strictly distinguishing t=0.5 and t=1 would require intraday data such as minute-level data (see Section 7).

### 3.7 SHAP Analysis

SHAP (SHapley Additive exPlanations) values are computed for each OOS prediction using shap.LinearExplainer with the interventional feature-perturbation scheme. SHAP is based on the Shapley value from cooperative game theory and decomposes each feature's contribution to a prediction. The interventional scheme replaces a feature's value with values randomly sampled from the marginal distribution of the training data, thereby controlling to some degree for dependence among features. This allows contributions to be compared even when strongly correlated variables such as VIX and SPX are present. However, SHAP decomposes contributions within the predictive model and does not identify causal relationships among variables; for highly correlated variables such as SPX and VIX in particular, the interpretation of contributions involves certain assumptions.

In each window, the explainer is initialized with the standardized training data and computes SHAP values for the test observation. The reported statistic is the mean absolute SHAP value (mean|SHAP|), interpreted as the average magnitude of each feature's contribution to the log-odds prediction. Positive SHAP values indicate a contribution toward an up move, and negative values toward a down move. This procedure is applied to all four markets in the cross-market analysis.

### 3.8 Evaluation Metrics and Baselines

The main metric is directional accuracy (DA). For T OOS samples, DA = (1/T) × Σ 1[sign($\hat{r}_t$) = sign($r_t$)] × 100 (%). Statistical significance is evaluated with a one-sample binomial Z-test against H0: DA = 50%. The Z-statistic is Z = (DA/100 − 0.5) / sqrt(0.25/n), with n = 522 OOS samples. Significance levels are *** ($p < 0.001$), ** ($p < 0.01$), and * ($p < 0.05$).

Three baselines are used. First, the random walk predicts up or down with 50% probability. Second, AR(1) predicts the next day's direction from the sign of the previous day's Nikkei 225 return. Third, the VIX-sign rule is a zero-parameter rule that predicts a rise in Asian markets the next day when VIX falls relative to the prior day. This rule embodies the simple logic that, based on the strong negative correlation between VIX and equity markets, a fall in VIX (i.e., improving

risk appetite) implies a rise in Asian markets. Because it has no model parameters, it carries no risk of overfitting to particular data and provides a natural benchmark for predictive performance.

This rule is positioned as a simple benchmark close to the interpretable upper bound of the logistic regression. It is not an established name in the prior literature but a comparison benchmark constructed in this study. Because it has no estimated parameters, it is less affected by overfitting from parameter estimation and serves as a control for checking whether predictive performance depends on a particular model specification.

# 4. Results

## 4.1 Main Results: Logistic Regression vs. LSTM

**Table 2: Comparison of main models**

| Model | DA (%) | Z | p | n | Sig. |
|---|---|---|---|---|---|
| Random walk | 50.00 | — | — | 522 | n.s. |
| AR(1) | 48.47 | -0.70 | 0.484 | 522 | n.s. |
| VIX-sign rule | 69.35 | 8.840 | <0.001 | 522 | *** |
| LSTM (6-seed average) | 64.51 | 6.50 | <0.001 | 502 | *** |
| Logistic regression | 67.82 | 8.141 | <0.001 | 522 | *** |

Logistic regression attains DA = 67.82% and the LSTM 64.51%. In this study's setting, a relatively simple linear model achieves high predictive accuracy. However, the statistical significance of the difference between the two models requires additional verification. The AR(1) baseline (48.47%) is statistically indistinguishable from a random walk ($Z = -0.70$, $p = 0.484$), confirming that the sign of the Nikkei 225's one-day-lagged return has no significant directional predictive power.

Whether the DA difference between logistic regression and the LSTM (3.31 points) is statistically significant should ideally be confirmed with a Diebold–Mariano test (Diebold & Mariano, 1995). This study interprets the result as circumstantial evidence that, at least under the feature set used here, no additional predictive improvement from a complex nonlinear model was observed.

Notably, the zero-parameter heuristic VIX-sign rule attains DA = 69.35%, comparable in accuracy to the logistic regression. Because VIX and SPX are strongly negatively correlated, the direction of VIX may proxy for the direction of the New York equity market. The carrier screening in Section 4.6 decomposes these contributions.

This result suggests that, at least under the feature set used here, high predictive performance can be obtained without introducing a complex nonlinear model. In this study's setting, no clear improvement from the LSTM's additional nonlinear capacity or sequential memory was observed.

## 4.2 Prediction-Horizon Analysis

**Table 3: Directional accuracy by prediction horizon**

| Horizon | DA (%) | Z | p | n | Sig. |
|---|---|---|---|---|---|
| h=1 (t-0.5) | 67.82 | 8.141 | <0.001 | 522 | *** |
| h=2 | 49.90 | -0.044 | 0.9651 | 521 | n.s. |
| h=3 | 52.12 | 0.965 | 0.3347 | 520 | n.s. |
| h=5 | 53.28 | 1.494 | 0.1352 | 518 | n.s. |
| h=10 | 50.10 | 0.044 | 0.9648 | 513 | n.s. |

The signal drops sharply from 67.82% at h=1 to 49.90% at h=2, statistically indistinguishable from a random walk ($Z = -0.044$, $p = 0.965$). This decay pattern is consistent with the semi-strong form of the efficient-market hypothesis (Fama, 1970). Predictability derived from New York close information declines rapidly within one business day, with no additional predictive power at longer horizons. That is, statistically significant predictability exists for the prediction problem corresponding to t=0.5, whereas none is confirmed from the next business day onward.

## 4.3 SHAP Analysis (Four-Variable Model)

**Table 4: Mean |SHAP| by market and feature**

| Market | MKT_lag | VIX | TNX | USDJPY |
|---|---|---|---|---|
| Nikkei 225 | 0.1725 | 0.4839 | 0.1254 | 0.0916 |
| KOSPI | 0.1078 | 0.3842 | 0.0384 | 0.0618 |
| Hang Seng | 0.1042 | 0.3450 | 0.0153 | 0.0476 |
| ASX 200 | 0.1247 | 0.5120 | 0.0176 | 0.0433 |

Table 4 reports mean|SHAP| by market and feature for the four-variable model. VIX is the dominant feature in all four markets, ranging from 0.345 (Hang Seng) to 0.512 (ASX 200). The market-specific differences in the interest-rate (TNX) and currency (USDJPY) channels are discussed in Section 5.2.

## 4.4 Sub-Period Robustness

**Table 5: Directional accuracy by sub-period**

| Period | Market environment | DA (%) | Z | p | n | Sig. |
|---|---|---|---|---|---|---|
| Full OOS (2021–2023) | — | 67.82 | 8.141 | <0.001 | 522 | *** |
| 2021 | Post-COVID recovery; ultra-low rates | 76.00 | 3.677 | 0.0002 | 50 | *** |
| 2022 | Rapid rate hikes; high VIX; bear market | 65.53 | 4.762 | <0.001 | 235 | *** |
| 2023 | Stabilization; soft-landing expectations | 68.35 | 5.651 | <0.001 | 237 | *** |

The signal is statistically significant at the 0.1% level in all three sub-periods. The lowest DA, in 2022 (65.53%), shows that the signal is maintained even amid the Fed's rapid rate hikes and a high-VIX environment—conditions combining high market volatility and a bear market. This suggests that t-0.5 predictability does not depend solely on a particular market environment. In other words, the results do not support the view that this predictability is limited to crisis-period cross-market transmission or depends only on low-volatility environments.

## 4.5 Cross-Market Analysis

**Table 6: Cross-market baseline comparison**

| Market | Random walk | AR(1) | VIX-sign rule | Logistic regression |
|---|---|---|---|---|
| Nikkei 225 | 50.00% n.s. | 48.47% n.s. | 69.35% *** | 67.82% *** |
| KOSPI | 50.00% n.s. | 50.63% n.s. | 60.58% *** | 60.94% *** |
| Hang Seng | 50.00% n.s. | 51.22% n.s. | 59.27% *** | 58.39% *** |
| ASX 200 | 50.00% n.s. | 49.54% n.s. | 68.65% *** | 67.28% *** |

Four main findings emerge. First, the t-0.5 signal is significant in all four Asia-Pacific markets (all $p < 0.001$), so it is not a Japan-specific phenomenon. Second, AR(1) is indistinguishable from a random walk in every market, so own-market momentum alone cannot explain the result. Third, the VIX-sign rule shows high accuracy in each market, indicating the importance of a common risk-appetite channel. Fourth, the performance of the logistic regression differs across markets, reflecting differences in market-specific financial and currency channels. DA levels differ across markets: 67–68% for the Nikkei 225 and ASX 200, about 61% for KOSPI, and about 58% for the Hang Seng. However, the statistical significance of these cross-market differences requires additional verification.

## 4.6 Carrier Screening: What Transmits the Signal

We expand the feature set from four to ten variables and re-run the walk-forward SHAP analysis for the Nikkei 225. The additional variables are SPX, TLT, GLD, DXY, OIL, and HYG, selected from five categories to evaluate which features contribute importantly to t-0.5 predictability: the U.S. equity market (SPX), safe-haven assets (TLT, GLD), global liquidity (DXY), commodity markets and risk-appetite indicators (OIL), and the credit market (HYG). All of these are observable by the New York close and are used as information available before the Tokyo open.

**Table 7: Carrier screening — mean |SHAP| ranking (Nikkei 225, n=519)**

| Rank | Variable | Mean \|SHAP\| | Channel | Description |
|---|---|---|---|---|
| 1 | SPX | 0.4310 | Equity | S&P 500 return — NY market direction |
| 2 | VIX | 0.2558 | Fear | Implied volatility — residual uncertainty |
| 3 | TLT | 0.1198 | Bond | Long-term U.S. Treasury ETF — safe-haven flight |
| 4 | GLD | 0.0601 | Safe haven | Gold ETF — risk-off barometer |
| 5 | DXY | 0.0425 | Currency | Dollar index — global liquidity |
| 6 | MKT_lag | 0.0416 | Own | Own-market one-day lag |

| | | | | |
|---|---|---|---|---|
| 7 | OIL | 0.0406 | Commodity | Crude oil futures — risk appetite |
| 8 | HYG | 0.0321 | Credit | High-yield bond ETF — credit spread |
| 9 | TNX | 0.0313 | Rates | 10-year U.S. Treasury yield |
| 10 | USDJPY | 0.0309 | Currency | USD/JPY exchange rate |

The expanded model attains DA = 70.33% ($Z = 9.262$, $p < 0.001$), exceeding the base four-variable model by +2.51 points, indicating that the additional variables contribute to improved predictive performance.

Three findings stand out. First, SPX is the largest carrier; the direction of the New York equity market itself is the strongest carrier. SPX shows the highest SHAP contribution (0.431), surpassing VIX (0.256). The high contribution of VIX observed in the base model may reflect its strong negative correlation with SPX; even after adding SPX, both SPX and VIX retain high contributions. Second, TLT ranks third (mean|SHAP| = 0.120), confirming a safe-haven channel operating through the bond market. Third, the contributions of MKT_lag, OIL, HYG, TNX, and USDJPY are relatively small (all below 0.043) and remain auxiliary information sources. Most of the contribution in the predictive model is concentrated in SPX, VIX, and TLT.

Overall, the signal observed in this study can be interpreted as being transmitted primarily by the NY equity-risk complex consisting of SPX, VIX, and TLT. Economically, SPX can be viewed as reflecting the direction of the New York equity market, VIX as investor sentiment and market uncertainty, and TLT as safe-haven demand. All three show high contributions in the SHAP analysis, and we refer to their combination, for convenience, as the NY equity-risk complex. This complex can be interpreted as a set of indicators summarizing information on equity markets, risk perception, and safe-haven demand observed at the New York close.

## 5. Discussion

### 5.1 Why Logistic Regression Outperforms the LSTM

The reason logistic regression outperformed the LSTM is likely attributable to sample size, the number of features, and the simplicity of the signal structure. Although the LSTM can capture nonlinearity and long-term dependence, the inputs here are a small number of daily market variables, and the main signal is concentrated in the direction of VIX and SPX. That logistic regression performed at least as well as the LSTM is one of the important methodological findings of this study. The LSTM's characteristic sequential-memory mechanism and nonlinear representational capacity did not lead to a clear improvement in predictive performance in this setting. This result does not deny the general usefulness of LSTMs; rather, it suggests that, at least under the feature set used here, a relatively simple linear model can achieve sufficient predictive performance. Indeed, the SHAP analysis shows that VIX has a high contribution, and the VIX-sign rule (69.35%) achieves accuracy comparable to the logistic regression. These results

are consistent with the possibility that a substantial part of predictability is explained by relatively simple information.

This has implications for the machine-learning literature in financial forecasting. Gu, Kelly, and Xiu (2020) demonstrated the effectiveness of neural networks—showing that they outperform linear models in cross-sectional prediction of U.S. equities, where many features interact in complex ways—but for short-horizon directional prediction that exploits time zones, an interpretable and reproducible linear model can be sufficiently competitive. By contrast, this study's t-0.5 prediction problem differs in nature in that predictive contributions are concentrated in a relatively small number of macro-financial variables. In such a setting, a complex nonlinear model may not necessarily yield additional predictive improvement.

## 5.2 The NY Equity-Risk Complex: Direction, Fear, and Safe Havens

The carrier-screening results show that information transmission to Asian markets is carried not by a single fear gauge alone but by composite risk information including the direction of the New York equity market, volatility, and U.S. long-term Treasuries. Specifically, three main groups of variables show high contributions in the predictive model.

### SPX (first)

The feature with the highest contribution was the S&P 500 return (mean|SHAP| = 0.431). This suggests that information about the direction of the New York equity market plays an important role in next-day Asian-market prediction. Indeed, a rise (fall) in the New York market tends to contribute in a direction consistent with predicting a rise (fall) in Asian markets the next day.

### VIX (second)

Even after SPX is included simultaneously, VIX retains a high contribution (mean|SHAP| = 0.256). This suggests that VIX reflects additional information about the market environment, rather than the direction of equity prices itself. For example, even when the S&P 500 falls by a similar amount, the uncertainty and risk aversion perceived by market participants may differ between episodes in which VIX rises sharply and those in which it does not. VIX can therefore be interpreted as an indicator reflecting the market's future uncertainty and risk perception, and this information may contribute to next-day Asian-market prediction.

This result is also consistent with why the zero-parameter VIX-sign rule (DA = 69.35%) achieved accuracy comparable to the logistic regression. Because VIX and SPX are strongly negatively correlated, the direction of VIX contains information closely related to the direction of the New York equity market; thus, VIX alone may capture a substantial part of predictability. At the same time, because VIX retains a high contribution even in the expanded model that explicitly includes SPX, it may contain additional information that cannot be explained by equity-price direction alone.

### TLT (third)

TLT shows the third-highest contribution (mean|SHAP| = 0.120). This suggests that information about the U.S. long-term Treasury market contributes to the predictive model from a dimension different from SPX and VIX. In general, a rise in TLT means a rise in Treasury prices and is often associated with increased demand for safe-haven assets. The result is therefore consistent

with the possibility that information about safe-haven demand contains useful information for next-day Asian equity-market prediction.

This result is also consistent with prior research on global portfolio rebalancing: developments in the U.S. bond market reflect information different from the equity market, and that information may be related to international asset-allocation behavior. While this study does not directly test that mechanism, the fact that TLT consistently shows a high contribution in the predictive model suggests that information originating in the safe-haven market is related to the directional prediction of Asian markets.

In the base four-variable cross-market analysis (Section 4.3), VIX appeared as the main explanatory variable, but when expanded to ten variables, SPX becomes the largest carrier. This suggests that VIX functions not in isolation but as part of a broader risk complex that includes the direction of the New York equity market.

On the other hand, in the Japanese market, the contribution of TNX in the four-variable model (0.125; Table 4) greatly exceeded that of the other markets (all below 0.04). This suggests that fluctuations in U.S. long-term interest rates may contain relatively important information for the Japanese market. The contribution of USDJPY is also relatively high (0.092; four-variable model), consistent with the possibility of strong linkages among U.S. interest rates, the exchange rate, and the Japanese equity market.

Economically, this result can be interpreted as consistent with mechanisms of capital flows and exchange-rate adjustment through the U.S.–Japan interest-rate differential. A rise in U.S. interest rates is often accompanied by yen-depreciation pressure, and its effect may be reflected in the predictability of the Japanese equity market. In Korea, Hong Kong, and Australia, by contrast, the contributions of TNX and USDJPY are relatively small, and similar linkages were not observed as clearly as in Japan.

## 5.3 Information Lifespan and Market Efficiency

The drop in predictive accuracy from 67.82% at h=1 to 49.90% at h=2 offers an interesting implication for market efficiency. The semi-strong form of the efficient-market hypothesis (Fama, 1970) predicts that public information is rapidly reflected in prices. The results show that predictability based on information available at the New York close is observed in the next business day's Asian market but is not confirmed from the following business day onward.

This result is consistent with New York market information being reflected in prices over a relatively short period. In particular, while statistically significant directional predictability exists at h=1, it declines to a level indistinguishable from a random walk at h=2; predictability decays rapidly within one business day.

That said, this study does not directly test the specific mechanism producing this decay. Possible factors include the non-synchronicity of trading hours across markets, differences in liquidity, and the rebalancing behavior of institutional investors, but verifying these would require additional analysis using order data or high-frequency data.

## 5.4 Regime Robustness and Contagion vs. Risk Premium

In the sub-period analysis, predictability was statistically significant in all three years—2021, 2022, and 2023 (all $p < 0.001$). This suggests that t-0.5 predictability is not a phenomenon limited to particular crisis periods. DA was highest in 2021 at 76.00% and relatively lower in 2022 at 65.53%, yet significance was maintained even in a market environment with rapid Fed rate hikes and high volatility. The results are therefore consistent with the possibility that predictability based on New York market information is observed not only in crisis periods but also under different market environments.

## 5.5 Economic Significance and Practical Implications

Directional accuracy of DA = 67.82% is more than mere statistical significance and is a non-negligible level in practice. However, taking into account transaction costs, slippage, short-selling constraints, and currency-hedging costs, this accuracy does not necessarily translate directly into investment returns.

Any such profitability estimate would be only a theoretical approximation and does not account for the relationship between prediction errors and the magnitude of returns.

There are, however, important caveats for actual trading implementation. First, liquidity is limited in the time before the Tokyo open, and slippage may occur. Second, what this study evaluates is directional accuracy; it does not consider the relationship between predictive success and the magnitude of returns. Third, large trades may generate market impact, reducing theoretical profitability.

More importantly, the aim of this study is not to develop a trading strategy but to understand the process of information transmission from the New York market to Asian markets. Predictability at t=0.5 is useful as an empirical indicator for assessing market efficiency, providing a framework for analyzing how much predictability persists, in which markets, and for how long. Verifying its practical usefulness as a trading strategy is left for future work.

## 5.6 Limitations

Several caveats accompany the interpretation of this study. Below we make explicit the main limitations relevant to the robustness of the results.

First, the prediction target is the sign of the Nikkei 225 close-to-close log return, which is largely governed by the opening gap (the difference between the previous close and the current open). Therefore, the t-0.5 predictability measured here may include both the predictability remaining at the open and the price change reflected through the opening gap. Separating the two requires additional verification that decomposes returns into the overnight gap (previous close → current open) and the intraday return (current open → current close) and compares directional accuracy for each (see Section 7).

Second, Nikkei 225 futures are traded during New York trading hours in the Osaka Exchange night session and on the CME, and may already incorporate a substantial amount of New York information before the Tokyo cash market opens. Part of the signal captured here may reflect the

price-discovery process by which the cash market follows the overnight futures price; additional analysis controlling for the overnight futures price is desirable.

Third, this study uses a binomial Z-test as a simple significance assessment, which assumes the independence of each prediction. Autocorrelation that may remain in the daily sign series, and multiple testing across four markets and several horizons, may act to overstate significance. Robustness checks using multiple-comparison corrections or block bootstrap are left for future work.

Fourth, the SHAP-value ranking indicates each variable's contribution within the estimated predictive model and does not directly indicate causal importance. The name NY equity-risk complex is a descriptive concept for organizing the group of variables that showed high contributions in the SHAP analysis, and does not assert the existence of any specific causal mechanism.

## 6. Conclusion

This study re-examined the international-information-transmission hypothesis of Lin, Engle, and Ito (1994) using the t-0.5 framework. Treating the New York close as the origin of information (t=0) and analyzing how much predictability is observed in the subsequent Asian market, we obtained the following five main findings.

First, significant directional predictability was confirmed at t=0.5. The directional accuracy of the Nikkei 225 from walk-forward logistic regression was 67.82% ($Z = 8.141$, $p < 0.001$), exceeding the random-walk benchmark of 50% by 17.82 points. This shows that information available at the New York close has significant predictive power for the next-day direction of Asian markets.

Second, SPX, VIX, and TLT were identified as especially important features in the predictive model. In the ten-variable SHAP analysis, the S&P 500 return (SPX) showed the highest contribution (mean|SHAP| = 0.431), followed by VIX (0.256) and TLT (0.120). The directional accuracy of the expanded model rose to 70.33%, suggesting that these variables provide additional information to the predictive model.

Third, predictability decayed rapidly within one business day. In the prediction-horizon analysis, directional accuracy at h=2 was 49.90%, statistically indistinguishable from a random walk. This indicates that predictability derived from New York market information disappears over a relatively short period, consistent with the semi-strong form of the efficient-market hypothesis.

Fourth, predictability was observed consistently across different market environments. In each sub-period—2021 (76.00%), 2022 (65.53%), and 2023 (68.35%)—statistically significant directional accuracy was confirmed, suggesting that the predictability observed here is not limited to a particular market phase.

Fifth, similar predictability was confirmed in the other Asia-Pacific markets analyzed. Statistically significant directional accuracy was obtained for KOSPI (60.94%), the Hang Seng

(58.39%), and ASX 200 (67.28%). While the importance of VIX is a common feature, the contributions of interest-rate and currency-related variables differed across markets.

This study showed that information at the New York close has significant predictive power for the next-day direction of Asian markets and clarified that SPX, VIX, and TLT play important roles in the predictive model. At the same time, what this study measures is directional predictability, not the amount of unpriced information itself; and the SHAP analysis indicates contributions within the predictive model rather than causal relationships. The results should therefore be interpreted as empirical evidence for the existence of predictability associated with international information transmission.

## 7. Directions for Future Research

This study showed that information at the New York close has significant predictive power for the next-day direction of Asian markets. However, it is an analysis using daily data and focusing on the single measurement point t=0.5; clarifying the full picture of the information-transmission process requires further research.

First, one direction for developing the framework is to capture the intensity of information transmission as a continuous function. The directional accuracy measured here indicates the existence of predictability exceeding the random-walk benchmark (50%), but it would be desirable to describe continuously how this predictability changes over time. If minute-level or tick data were available, predictability could be estimated at higher temporal resolution, allowing detailed analysis of the speed at which predictability derived from New York close information decays over the overnight period. This study is a first step, using daily data and focusing on the discrete measurement point corresponding to t=0.5.

Relatedly, setting multiple measurement points within the 24-hour trading cycle is also an important task. This study analyzed a single measurement point corresponding to the Tokyo open, but using the open and close of each market as observation points could depict the temporal profile of information transmission in greater detail. For example, taking the open and close of the New York, London, and Tokyo markets as measurement points would provide multiple observation points within the 24-hour cycle. This would make it possible to examine how much predictability derived from New York close information decays by the opening of the London market, and whether the decay speed differs between transmission via London and direct transmission to Asian markets.

Furthermore, the t-0.5 framework proposed here can be readily applied to other market pairs. For example, comparing information transmission across markets such as New York→London, London→Tokyo, and New York→Shanghai could map the global distribution of cross-time-zone predictability. While consistent predictability was confirmed in four Asia-Pacific markets here, it would be of interest to test whether similar patterns are observed in European markets and whether VIX continues to function as an important predictor.

Extending the analysis from the overall market to industry- and sector-level analyses would also be valuable. Decomposing the predictability observed at the market-index level by industry could

reveal which aspects of New York market information are transmitted faster or slower to particular sectors. For example, one could test whether information-transmission speed differs between the technology and financial sectors, and whether VIX or SPX is the main predictor.

In addition, understanding the economic nature of the observed predictability more deeply requires clarifying its relationship with risk premia. For example, regressing the returns of a predictive strategy on a multi-factor model including Fama-French factors and a global volatility factor could distinguish whether the observed predictability is due to alpha (pricing distortions) or beta (a structural risk premium). This is also important for understanding whether the predictability observed here is unlikely to be eliminated by arbitrage.

The use of high-frequency data is particularly promising. If minute-level data become available, the temporal evolution of predictability could be estimated continuously, and the point after the New York close at which each variable is most strongly reflected in prices could be analyzed more precisely, allowing the international information-transmission process to be described at higher resolution.

Finally, because the prediction target of this study is the sign of the close-to-close return, a limitation is that the contributions of the opening gap and the intraday return are mixed. Future work should separate the overnight gap (previous close → current open) from the intraday return (current open → current close) and measure directional accuracy for each. This would make it possible to more clearly identify whether New York close information is reflected in prices mainly at the open or continues to be incorporated after the open. The opening price required for this decomposition is included in standard daily data (e.g., Yahoo Finance), so this verification can be carried out at no additional data cost.

Through these extensions, the t-0.5 framework proposed in this study is expected to provide a foundation for further developing research on information transmission and predictability in international financial markets.

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used generative AI tools (large language models) to translate the manuscript from Japanese into English and to improve the readability and language of the text. These tools were not used to generate research insights, design the analysis, analyze or interpret the data, or draw conclusions. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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